

ASSIGNMENT #2

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
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Due Date: Friday February 7 by 5:00pm in my office

Number of points.....100
Your score.....80

Note: All problems are worth 100 points. Your final score will be the average value

Preface

In this assignment the Regenerative Rankine Cycle will be analyzed. How the analysis was carried out was first, going through the cycle and analyzing each piece of equipment carrying out an energy balance, first under ideal conditions (assumptions will be given later), then using the results from the ideal conditions I used those to calculate actual values. I had used the CHE301 calculator to carry out my analysis and aid with my computations.

Energy Balances/assumptions

There were five pieces of equipment analyzed before data was gathered. The turbine, pump, mixer, condenser, and boiler. It was helpful to carry out an energy balance on all the equipment first before doing any calculations. Using the formula for the energy balance $\Delta H = \dot{Q} + \dot{W}$ eq 1.0 where ΔH is the enthalpy, $\dot{Q}$ is the rate of heat transfer in kJ/s and the $\dot{W}$ is the rate of work in kJ/s (-1)

Turbines HPTURB and LPTURB

The first piece of equipment analyzed was the low pressure and high pressure turbines. The pressure in and out does not affect the equation of the energy balance so this stays consistent across both turbines. We assume that heat transfer is negligible so the equation that we arrive at using eq 1.0 is

$$\Delta H \dot{m} = \dot{W} \text{ eq 1.1 } \checkmark$$

This energy balance was used for both ideal and real cases since one of the assumptions stated was that the rate of heat loss was negligible. But to solve for real enthalpy an equation must be derived for the real entropy of a turbine. Starting with the equation 1.1 we may be able to solve for our real turbine enthalpy.

$$w_i = (h_1 - h_2) * \dot{m}$$

It is known also that the real work is equal to the efficiency of the turbine times the ideal work

$$N w^i = w^r$$

Substituting enthalpies we arrive at this final equation for the real enthalpy, where N is the efficiency and the h 's are the enthalpy.

$$H_r = N(h_2 - h_1)_i + h_1 \text{ eq 1.2 } \checkmark$$

Ideal vs real

For the ideal case we assume 100% efficiency of the turbine vs. the 88% that was given to us. Using the ideal assumption we also know that the entropy in the inlet and outlet of the turbine is isentropic. This is a characteristic of the ideal turbine. $\checkmark$

$$-\Delta S + \frac{\dot{Q}_0}{T} + \frac{\dot{Q}_1}{T} + \dot{S}_{gen} = 0$$

Since we assume no heat transfer (adiabatic), $\checkmark$ and we assume reversibility so $\dot{S}_{gen} = 0$ the entropy balance is

$$\Delta S = 0 \quad \checkmark$$

For the irreversible/real process we know that there is entropy generation so the entropy balance will include the generation term. $\checkmark$

$$-\Delta S + \dot{S}_{gen} = 0$$

Condenser

Through the condenser not much really will change from reversible to irreversible process, the energy balance and assumptions will stay the same but the rates of heat transfer will change. Due to changes in enthalpy from other streams. $\checkmark$ So the energy balance is as follows. Since the condenser does no work on the fluid, and it only removes the latent heat via the cooling water. We use eq 1.0 to arrive at our energy balance for the compressor.

$$\Delta H \dot{m} = \dot{Q} \text{ eq 1.3} \quad \checkmark$$

Lpump and Hpump

Like the turbine the pressures do not affect the terms in initial equations for the energy balances. Similarly we assume that the pump has negligible heat transfer, $\checkmark$ but the term for work stays so we have the same equation as for the turbine. The pump will do work on the fluid going through it and thus the term for work must stay in the equation.

$$\Delta H \dot{m} = \dot{W} \text{ eq 1.1}$$

The equation for the real enthalpy of solution can be determined by a derivation. Using the equation for the ideal pump:

$$W = \Delta H$$

And that the definition of work is pressure times a differential of volume we get the equation

$$(P_2 v_2 - P_1 v_1) = \Delta H$$

But since for a pump the pressure is the only thing changes $v_1=v_2$ so it may be factored out

$$v(P_2 - P_1) = \Delta H \quad \checkmark$$

Substitutions can be made and rearranging of the equation gets us the equation where N is the efficiency and h is the enthalpy.

$$H_{2r} = \frac{(h_2 - h_1)_i}{N} + h_{1r} \text{ eq 1.4}$$

Mixer

The mixer's balance does not change from ideal to real streams. Then since there is no work and no heat transfer, we only have enthalpy to consider. So the energy balance is as follows, this holds for the ideal and real cases.

$$\Delta H = 0 \text{ eq 1.5} \quad \checkmark$$

Boiler

The boiler is the last piece of equipment that must be analyzed in the cycle. The energy balance for the boiler is as follows, since there is work being done on the water we can assume that the rate of heat transfer is equal to the enthalpy. This does not change from the ideal to real case.

$$\Delta H \dot{m} = \dot{Q} \text{ eq 1.6}$$

IMPORTANT CONVERSION

$$1,000 * 10^6 \frac{\text{ft}^3}{\text{day}} \cdot \frac{1 \text{ day}}{86400 \text{ s}} * \frac{14 \text{ psia}}{(10.7239 \frac{\text{psia ft}^3}{\text{lbmol}}) * 519.67 \text{ R}} * \frac{29.16 \text{ lbmol}}{2.204 \text{ lbmol}} * \frac{1 \text{ kmol}}{2.204 \text{ lbmol}} = 13.889 \frac{\text{kmol}}{\text{s}}$$

Why? (-1)

S2 ideal, bypass ideal, S3 ideal

Molar Composition	Units	S1 Ideal	S2 Ideal	S Bypass Ideal	Real Bypass	S2 Irreversible	S3 Ideal
100	mol fraction						
Conditions and Properties							
Units							
Molar Flow	kmol/s	13.8440	13.8440	3.2643	2.7532	13.8440	19.0000
Mass Flow	kg/s						
Pressure	bar	80.0000	7.0000	7.0000	7.0000	7.0000	7.0000
Temperature	°C	480.0000	164.0000	164.0000	164.0000	164.0000	164.0000
Volume Fraction (Quality)							
Molar Enthalpy	kJ/kmol	19.0000	18.0000	18.0000	18.0000	18.0000	18.0000
Molar Entropy	kJ/kmol	19.0000	18.0000	18.0000	18.0000	18.0000	18.0000
Molar Density	kmol/m³						
Enthalpy Flow	kJ/s						
Entropy Flow	kJ/s						

This analysis was carried out using the assumptions that outlined in the previous section. Stream 1 is given to us nothing much more to be said here. Stream 2, is the first stream that come out equipment, this stream comes out of the ideal turbine. The steam enters as a superheat vapor into the turbine at 480°C and 80 bar and flow rate of 1000 MMSCFD (conversion) this is 13.844 kmol/s. This exits the turbine at 7 bar. Since the turbine is ideal we can assume 100% efficiency so we look at the steam tables, to extrapolate data. From the table we know that the temperature of stream 2 ideal will be the saturated

Figure 1

temperature since the ideal turbine operates at 100% efficiency. Using the data from the steam tables and the functions of the 301 calculator it is determined that the vapor fraction of the stream coming out of the ideal turbine is 99.08% vapor. This was determined by the entropy since from the steam tables we were able to obtain the liquid and vapor entropies. We used these values coupled with the lever rule equation to determine the quality of the stream 2.

This was working off the assumption that the process was isentropic so using the extended secant function on the 301 calculator the quality was obtained by setting the two entropies equal. The work of the turbine was determined with the equation 1.1 and multiplying by the molar flow rate of 13.884 kmol/s the max work of the turbine is -151197.46 kJ/s. This is consistent with a turbine having a power output. The bypass and stream 3 retain the qualities, and properties of stream 2. The only thing that will change between the streams is the flow rates of the streams, which can be hypothesized reasonably that the stream 3 will have a bigger flow rate than the flow of the bypass since the turbine will require more liquid than the bypass. ? (-1)

S2, S3 and bypass Irreversible

The actual case was not as convenient as the ideal case that we can set the two entropies to each other. This time it also can not be assumed that the steam exits as a saturated vapor/liquid solution. First thing is first, the irreversible enthalpy can be solved for using the equation 1.2 we get are able to solve for enthalpy. This enthalpy is used to solve for the temperature of the stream, by the 301 calculator we can obtain the temperature in degrees Celsius. The method to obtain this was using the

S4,S5,S6 Ideal

[illegible]

we know the temp and pressure so we solve for enthalpy luckily also no phase change happens, so the quality remains 0.00.

S4 out of the LPTURB is a liquid vapor mixture. This is apparent for ideal and real cases. Using equation 1.2 we obtain the real enthalpy, and taking the quality and temperature into account we use the solver tool in excel to calculate the quality and temperature of the solution, by setting the enthalpy calculated to the HSteam function and the temp Cell by changing the quality. Stream 5 retained qualities of S4 but we know that through the condenser that the latent heat is removed so the solution coming out of the condenser is 100% liquid. Using equation 1.4 we get the real enthalpy of stream 6 and work is solved for by using the equation 1.1.

This is where we can solve for the flow rate through the bypass and the S3,4,5,6. This is accomplished by setting up a mass balance, and the energy balance. Where n_b is the molar flow rate of the by pass

$$(-1)^{?} \circ 1 = n_b + n_6$$

$$\dot{n}_7 H_7 = \dot{n}_b H_b + \dot{n}_6 H_6$$

3. The efficiency of the cycle

The efficacy of a power cycle is the work over heat into the system $\frac{W}{Q_H}$, this efficiency is lower than the ideal efficiency which checks out. The ideal efficiency was calculated to be at 45% so the drop off was about 6% so this is making sense.

$$\text{abs}(-126757.6 \text{ Kw} + -122579 \text{ Kw} + 148.51 \text{ kw} + 1234.47 \text{ kw}) / 628578.6110 \text{ kw} =$$

39.3%

x

(-2)

4. The amount of water in **gpm** required for the condenser, assuming a water temperature rise of 11°C

The amount of water in Gallons per minute required for the condenser is going to be $\frac{Q_{out}}{C_p \Delta T}$ so substituting into the equation will be $\frac{380153 \frac{\text{kJ}}{\text{s}}}{4.11 \frac{\text{kJ}}{\text{kg}^\circ\text{C}} \cdot 11^\circ\text{C}}$ converting to GPM arriving at the answer

130289.7 gpm

x

at what temp?

5. If the power plant is using coal as fuel in the boiler give an estimate of the amount of coal needs in **tonne/day**

*reference? (-2)

Looking up the heating value of coal we are able find out how many tons/day the power plant needs the heating value of coal is between 25-20 Mega Joules/kg so I used about half way 23.9 MJ/kg so dividing the heating value by the heat produced by the boiler in Mega Joules and then taking the inverse we would solve for the Kilograms needed per second of coal but the answer should be in tonnes(metric) per day, arriving at the value for coal this is a lot of coal, but this answer does make sense since the amount of electricity is a lot.

$$\frac{23.9 \frac{\text{MJ}}{\text{kg}}}{661.706 \frac{\text{MJ}}{\text{s}}} = 26.2 \frac{\text{kg}}{\text{s}} \cdot \frac{86400 \text{ s}}{\text{day}} \cdot \frac{0.001 \text{ tonnes}}{1 \text{ kg}} = 2392.1 \text{ tonnes/day}$$

needs in tonne/day

6. If the power plant is equipped with cooling tower, how much water does it draw from the environment, and how much it discharges as waste. Express your answers in **gpm**

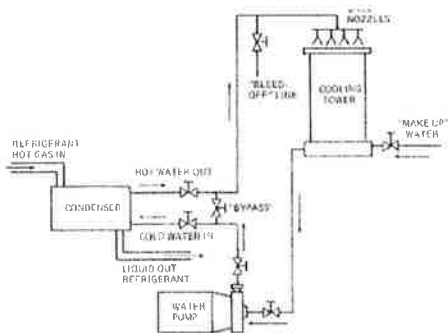


Figure 8-14 Retreating water system using a tower

focusing on the cooling tower in this graphic below we see that the cooling tower has two ways to get rid of water. The stream to the water pump and the evaporation but it also has two ways to regain the water loss which is the make up water and the hot water. Assuming that the wind drift is 0. We can set up a mass balance. On the cooling tower. Where:

$$\text{Make-up} = \text{Evaporation} + \text{Draw} \quad \checkmark$$

reference?

Additionally we can set up a molecular balance on the power plant. This can be done with the minerals in the water, this can be done as follows.

$$x_m \text{Make} - \text{up} = x_d \text{Draw} \checkmark$$

Where we also know the ratio of minerals in the Draw water to the make up water is ^{reference? (-2)} 5 to 1 solving for one of the variables we get this equation

$$4D = E$$

Using the equation for the evaporation as

$$E = \frac{Cp\Delta T * \text{flow}}{\Delta H_{vap}} \quad \text{where this eqn comes from? (-1)}$$

Solving using the flow rate in the last problem gets us

at what temp.?

$$\text{Using Evaporation} = \frac{4.11 \frac{\text{kJ}}{\text{kg } ^\circ\text{C}} * 11 ^\circ\text{C} * 1302289 \text{ GPM}}{2424 \frac{\text{kJ}}{\text{kg}}} = 2430.03 \text{ gpm}$$

at what temp. ? (-2)

Then if we plug this into the equation above we get

Draw= 607 gpm

the make up is 3037.53 gpm

